SilverStar Navigation Models

Pure INS, SF6, KF6 and ESKF15

Implementation reference for SilverStar 0.1.2

Source baseline:
`234caa5f6d8e704739edb897078666274fca0673`

1 Scope and version identities

This reference describes the implemented firmware equations and their recorded-data boundaries. It is based on source inspection at the commit printed on the title page. It does not establish sensor accuracy, target execution time, stack safety, electrical correctness or flight qualification.

The product version is 0.1.2. Independent identities remain AIR M0, SSLOG 0.0 and `.ssdecoder` package schema/project semantics 1.2. Algorithm revisions are separate: SF6 revision 2, KF6 process model revision 2, ESKF15 revision 1 and navigation-quality revision 3. A historical log retains the exact revision and decoder with which it was recorded. A current mathematical description cannot replace those identities.

These algorithms share navigation coordinates, but their stored states differ:

Algorithm	State and observation feedback	Covariance
Pure INS	Independent inertial $\boldsymbol{p}, \boldsymbol{v}, q_{nb}$; no observation correction	None
SF6	Six velocity/position states; fixed per-axis correction; separate inertial attitude	None
KF6	Six position/velocity states; GNSS/barometer correction; external software attitude	6×6
ESKF15	Nominal $\boldsymbol{p}, \boldsymbol{v}, q_{nb}, \boldsymbol{b}_g, \boldsymbol{b}_a$; right local error feedback	15×15

Feature tier and hardware feasibility remain configuration contracts. Full exposes the supported estimator choices; Standard excludes ESKF15; Compact/Minimal permit SF6 or pure INS and exclude KF6/ESKF15. A supported choice is not hardware acceptance. The independent Pure INS branch is not overwritten by estimator corrections.

References below use source keys S1–S14, resolved in Section 10; each key names an owning repository file and functions. No external project path or machine-local build directory is required to maintain this document.

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2 Coordinates, symbols and units

The local navigation frame n is East–North–Up (ENU). Body frame b follows the selected mounting-axis mapping. Vectors are columns and are ordered E,N,U when in n. Local position is measured from a frozen GNSS origin, not directly in latitude/longitude units. Up-positive gravity is

$$\boldsymbol{g}_n = \begin{bmatrix} 0 & 0 & -g \end{bmatrix}^T, \qquad g > 0. \tag{1}$$

Accelerometer input is *specific force*: a stationary, level sensor supplies approximately $+g$ on its up axis; $R_{nb}\boldsymbol{f}_b + \boldsymbol{g}_n$ is navigation acceleration. Subtracting gravity in Body before an arbitrary rotation would change this model.

Symbol	Meaning	Unit
$\boldsymbol{p}, \boldsymbol{v}$	Local ENU position and velocity	m, m/s
$\boldsymbol{f}_m, \boldsymbol{\omega}_m$	Body specific force/angular rate after external calibration	m/s^2, rad/s
$\boldsymbol{f}, \boldsymbol{\omega}$	ESKF input after subtracting estimated residual biases	m/s^2, rad/s
q_{nb}, R_{nb}	Hamilton scalar-first quaternion $[w, x, y, z]^T$ and Body-to-ENU rotation	dimensionless
$\Delta\boldsymbol{\theta}_b$	Integrated Body rotation vector	rad
$\Delta\boldsymbol{u}_b$	Integrated specific force before gravity; not ENU velocity	m/s
$\Delta\boldsymbol{v}_n$	Rotation- and gravity-corrected navigation increment	m/s
h, h_i, T	Current integration duration, subinterval duration, original unsplit KF interval	s
t_s, t_r, t_e	Measurement/sample, receive and evaluation timestamps	s; stored as μs
$\boldsymbol{b}_g, \boldsymbol{b}_a$	ESKF residual gyro and accelerometer bias	rad/s, m/s^2
$\delta\boldsymbol{x}$	ESKF position, velocity, right Body attitude and bias errors	mixed state units
P, Q_d	State/error covariance and discrete process covariance	pairwise state-unit products
R, S	Observation covariance and innovation covariance	pairwise observation-unit products
H, K	Observation Jacobian and gain	corresponding state/observation ratios
$\boldsymbol{r}, \eta$	Innovation and normalized innovation squared (NIS)	observation units, dimensionless
$\boldsymbol{\ell}$	Lever arm from state reference to GNSS antenna, in Body	m
σ_a	KF6 per-original-interval acceleration standard deviation parameter	m/s^2
n_g, n_a	ESKF gyro/accelerometer amplitude density	rad/s/$\sqrt{\text{Hz}}$, m/s^2/$\sqrt{\text{Hz}}$
n_{bg}, n_{ba}	ESKF bias derivative amplitude density	rad/s^2/$\sqrt{\text{Hz}}$, m/s^3/$\sqrt{\text{Hz}}$

Hamilton multiplication, with $q = [w, \boldsymbol{u}]^T$ and $d = [s, \boldsymbol{a}]^T$, is

$$q \otimes d = \begin{bmatrix} ws - \boldsymbol{u}^T\boldsymbol{a} \\ w\boldsymbol{a} + s\boldsymbol{u} + \boldsymbol{u} \times \boldsymbol{a} \end{bmatrix}. \tag{2}$$

The rotation used by the implementation is

$$R(q) = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{bmatrix}. \tag{3}$$

Thus $\boldsymbol{u}_n = R(q_{nb})\boldsymbol{u}_b$. Body increments multiply on the right. The skew convention is $[\boldsymbol{a}]_\times \boldsymbol{b} = \boldsymbol{a} \times \boldsymbol{b}$.

For $\alpha = \|\boldsymbol{\theta}\|$, the quaternion exponential is

$$\mathrm{Exp}(\boldsymbol{\theta}) = \begin{bmatrix} \cos(\alpha/2) \\ \dfrac{\sin(\alpha/2)}{\alpha}\boldsymbol{\theta} \end{bmatrix}, \tag{4}$$

with its continuous zero-angle limit. Pure INS uses $1 - \alpha^2/8$ and $1/2 - \alpha^2/48$ when $\alpha^2 \le 10^{-6}$, then normalizes; ESKF uses $\cos(\alpha/2)$ and the small-angle vector factor $1/2 - \alpha^2/48$ for $\alpha < 10^{-3}$. They implement the same convention with slightly different numerical branches. Quaternion normalization enforces a valid rotation; it does not remove attitude drift. q and $-q$ represent the same orientation.

Quaternion/Euler ENU yaw is $\mathrm{atan2}(R_{21}, R_{11})$ for the Body $+X$ axis projected horizontally; zero points east. A known-yaw alignment target can instead constrain a selected signed Body reference axis, as specified in Section 7. North-referenced compass bearing must not be substituted without an explicit convention conversion.

Implementation: S2 and S6, quaternion multiplication, rotation, propagation and injection.

3 Pure INS: coning/sculling mechanization

Pure INS maintains its own attitude, velocity and position from the admitted initial state. Its calibration-corrected Body input is independent of estimator feedback. It uses a local flat frame with constant configured gravity; Earth rotation, transport rate and Coriolis acceleration do not appear in the implemented propagation.

For successive samples, the actual timestamp difference gives each duration h_i. A trapezoidal subinterval supplies

$$\boldsymbol{a}_i = \tfrac{1}{2}(\boldsymbol{\omega}_{i-1} + \boldsymbol{\omega}_i)h_i, \qquad \boldsymbol{u}_i = \tfrac{1}{2}(\boldsymbol{f}_{i-1} + \boldsymbol{f}_i)h_i. \tag{5}$$

Two admitted subintervals give the implemented second-order corrections

$$\Delta\boldsymbol{\theta}_b = \boldsymbol{a}_1 + \boldsymbol{a}_2 + \tfrac{2}{3}(\boldsymbol{a}_1 \times \boldsymbol{a}_2), \tag{6}$$

$$\Delta\boldsymbol{u}_b = \boldsymbol{u}_1 + \boldsymbol{u}_2 + \tfrac{1}{2}[(\boldsymbol{a}_1 + \boldsymbol{a}_2) \times (\boldsymbol{u}_1 + \boldsymbol{u}_2)] + \tfrac{2}{3}(\boldsymbol{a}_1 \times \boldsymbol{u}_2 + \boldsymbol{u}_1 \times \boldsymbol{a}_2). \tag{7}$$

The half coefficient handles rotation of the specific-force increment; the two-thirds terms handle the pair's coning/sculling correction. These coefficients describe the algorithm being used, not a guarantee for arbitrary sensor timing or motion.

With $h = h_1 + h_2$, the *starting* attitude rotates the corrected increment once:

$$\Delta\boldsymbol{v}_n = R(q_k)\Delta\boldsymbol{u}_b + \boldsymbol{g}_n h, \tag{8}$$

$$\boldsymbol{v}_{k+1} = \boldsymbol{v}_k + \Delta\boldsymbol{v}_n, \tag{9}$$

$$\boldsymbol{p}_{k+1} = \boldsymbol{p}_k + \tfrac{1}{2}(\boldsymbol{v}_k + \boldsymbol{v}_{k+1})h, \tag{10}$$

$$q_{k+1} = \mathrm{normalize}(q_k \otimes \mathrm{Exp}(\Delta\boldsymbol{\theta}_b)). \tag{11}$$

Diagnostic navigation acceleration is $\Delta\boldsymbol{v}_n/h$. Basic, rotation-corrected and sculling-corrected increments are separate recorded quantities; they are not three independent physical observations.

Firmware also accepts the configured one-subinterval mode by setting the second pair to zero. The current FLP Corrected-IMU Pure INS/KF6 replay requires the two-subinterval, undecimated

input contract; a valid recorded firmware output does not automatically supply that replay input. Formula validity and replay availability must be distinguished.

Admission checks include finite accelerometer/gyro input, valid flags, strictly advancing timestamps, and the duration range defined by the configured mechanization sample-rate limits and tolerance. A gap resets sample-pair collection and records a health failure. A rejected interval must not be replaced with a guessed uniform duration. Candidate navigation state and quaternion must pass the implemented finite/normalization checks before publication.

Implementation: S1, subinterval increments, coning/sculling, navigation transform and candidate publication.

4 SF6 revision 2: fixed-gain current-state fusion

SF6's state order is

$$\boldsymbol{x}_s = [v_E, v_N, v_U, p_E, p_N, p_U]^T. \tag{12}$$

It uses the admitted inertial navigation increment to predict

$$\boldsymbol{v}_{k+1}^- = \boldsymbol{v}_k^+ + \Delta\boldsymbol{v}_n, \tag{13}$$

$$\boldsymbol{p}_{k+1}^- = \boldsymbol{p}_k^+ + (\boldsymbol{v}_k^+ + \tfrac{1}{2}\Delta\boldsymbol{v}_n)h. \tag{14}$$

The separate attitude propagates inertially and is not corrected by these observations. For each observed state component j, the update is

$$x_{s,j}^+ = x_{s,j}^- + k_j(z_j - x_{s,j}^-), \qquad 0 \le k_j \le 1. \tag{15}$$

The six gains follow the state order above; their declared defaults are all 0.2. This is an engineering default, not an optimal gain or accuracy result. $k_j = 0$ retains the prediction and $k_j = 1$ selects the admitted observation. SF6 maintains no P, Q, R, Kalman gain or NIS. The shared output's statistical fields are unavailable (NaN), not zero uncertainty.

Group	Observation	Corrected components
0	GNSS position EN	p_E, p_N
1	GNSS position U	p_U
2	GNSS velocity EN	v_E, v_N
3	GNSS velocity U	v_U
4	Relative barometer altitude	p_U

GNSS altitude and barometer altitude share the p_U gain. Position observations do not directly correct velocity. There is no magnetic measurement update in SF6.

Revision 2 applies every admitted observation to the *present* predicted state. It has no history ring, delayed observation queue, rewind or replay. Measurement time serves admission/age checks. The backend uses a trusted sample timestamp, otherwise the receive timestamp (zero configured delay). A measurement must satisfy

$$t_0 \le t_s \le t_r \le t_e, \qquad t_e - t_s \le 0.5 \text{ s}. \tag{16}$$

For a previously seen group, both sample and receive times must advance. Unsigned 32-bit sequence difference must lie in $[1, 2^{31} - 1]$, which admits wraparound only within the forward half range. Prediction requires $0 < h \le 0.02$ s and exact continuity with the microsecond-rounded duration.

A constant-velocity delayed position observation lags current position by about $-\boldsymbol{v}\tau$. Equation (15) therefore adds approximately $-k_j v_j \tau$ beyond a current observation's correction. Under constant acceleration, delayed velocity analogously contributes $-k_j a_j \tau$. These are explanatory model estimates, not measured navigation accuracy.

The backend checks source identity, basic physical GNSS/barometer validity and age. SF6 does not obtain covariance-based satellite/window/NIS downweighting simply because those policies exist for KF6/ESKF15. Its health reports admission and whether the relevant gain can make an effective update; a zero gain cannot refresh successful-fusion health. The covariance-free core commits the finite candidate state and group admission metadata together.

Historical SF6 revision 1 used fixed-lag replay. Recorded mode must retain that revision's semantics; current Offline What-if revision 2 must identify its semantic difference. Continuing inertial propagation or barometer correction during GNSS loss does not bound horizontal drift.

Implementation: S3, fixed gains, state advancement, group/time admission and candidate commit. Implementation: S4, physical admission, current-state measurement timestamps and health/statistical output.

5 KF6 revision 2: position/velocity Kalman filter

The implemented state order differs from SF6:

$$\boldsymbol{x}_k = [p_E, p_N, p_U, v_E, v_N, v_U]^T. \tag{17}$$

Attitude propagation belongs to the software INS branch. KF6 does not estimate attitude or IMU biases. An already-rotated, gravity-corrected navigation increment is its prediction input.

5.1 Prediction and the original-interval process model

$$A(h) = \begin{bmatrix} I_3 & hI_3 \\ 0 & I_3 \end{bmatrix}, \qquad B(h) = \begin{bmatrix} \frac{1}{2}hI_3 \\ I_3 \end{bmatrix}, \tag{18}$$

$$\boldsymbol{x}^- = A(h)\boldsymbol{x}^+ + B(h)\Delta\boldsymbol{v}_n, \qquad P^- = A(h)P^+A(h)^T + Q_d(h;T). \tag{19}$$

Here T is the original, unsplit inertial interval and $h \le T$ is the fragment being propagated. For each axis i, the actual parameter is acceleration standard deviation $\sigma_{a,i}$ in $\mathrm{m/s^2}$. Revision 2 forms $d_i = \sigma_{a,i}^2 T$, giving

$$Q_{d,i}(h;T) = d_i \begin{bmatrix} h^3/3 & h^2/2 \\ h^2/2 & h \end{bmatrix} \quad \text{in the } [p_i, v_i] \text{ subspace.} \tag{20}$$

The full matrix inserts these three blocks at $(i, i+3)$ without cross-axis process noise. The units are $\mathrm{m^2}$, $\mathrm{m^2/s}$ and $\mathrm{m^2/s^2}$. The complete interval uses $h = T$.

This is the integral of an equivalent white-acceleration model for that original interval. Splitting an interval preserves T in every fragment:

$$Q_d(h_1 + h_2; T) = A(h_2)Q_d(h_1;T)A(h_2)^T + Q_d(h_2;T). \tag{21}$$

Replacing T by each fragment's duration breaks this noise accumulation. It is also incorrect to interpret the configured σ_a as the ESKF continuous amplitude-density parameter.

Historical revision 1 used $\mathrm{Var}(\Delta v_i) = (\sigma_{a,i}h)^2$ and $G = [h/2, 1]^T$, hence $Q_i = G\,\mathrm{Var}(\Delta v_i)G^T$. This historical model must remain available for its recorded decoder/revision; it must not be silently replaced with Equation (20).

5.2 Actual observations and units

With axis-selection rows $E_{EN} = [\boldsymbol{e}_E^T; \boldsymbol{e}_N^T]$ and $E_U = \boldsymbol{e}_U^T$, the production group matrices are

$$H_{p,EN} = [E_{EN} \;\; 0], \qquad H_{p,U} = [E_U \;\; 0], \tag{22}$$

$$H_{v,EN} = [0 \;\; E_{EN}], \qquad H_{v,U} = [0 \;\; E_U], \tag{23}$$

$$H_{baro} = [E_U \;\; 0]. \tag{24}$$

Groups have independent validity and updates. Do not append a second 3D GNSS update to observations already consumed through horizontal/vertical groups. Generic 3D APIs and thresholds exist, but do not describe an extra production observation.

Receiver-reported uncertainty is first scaled and floored as a standard deviation, then squared. For the generated actual profile,

$$\sigma_{p,EN} = \max(1.25\, hAcc, \sigma_{p,EN}^{floor}), \tag{25}$$

$$\sigma_{p,U} = \max(1.25\, vAcc, \sigma_{p,U}^{floor}), \tag{26}$$

$$\sigma_{v,EN} = \max(1.25\, sAcc, \sigma_{v}^{floor}), \tag{27}$$

$$\sigma_{v,U} = c_U \max(1.25\, sAcc, \sigma_{v}^{floor}). \tag{28}$$

The declared current defaults are floors 1.5 m, 2.5 m and 0.15 m/s, with $c_U = 1.75$. The effective group R contains variances and any admitted quality multipliers from Section 8. Invalid or missing supported-field evidence is not fabricated uncertainty. Barometer observation is relative up altitude with $R_{baro} = \sigma_{baro}^2$; its declared default sigma is 2.5 m.

Configured P_0 diagonal values already are variances and are not squared again. Declared floors are $(4, 4, 9)$ m^2 for position and $(0.25, 0.25, 0.25)$ m^2/s^2 for velocity. Qualified initial receiver uncertainty may increase applicable entries. Initial position is zero in the frozen local frame; initial velocity comes from the admitted initial-state snapshot. Source-specific validity still determines which components are usable.

5.3 Innovation, gates and covariance update

$$\boldsymbol{r} = \boldsymbol{z} - H\boldsymbol{x}^-, \qquad S = HP^-H^T + R, \tag{29}$$

$$\eta = \boldsymbol{r}^T S^{-1} \boldsymbol{r}, \qquad K = P^-H^T S^{-1}, \tag{30}$$

$$\boldsymbol{x}^+ = \boldsymbol{x}^- + K\boldsymbol{r}. \tag{31}$$

KF6 solves the innovation system using Cholesky-based routines. For dimension m, $\eta \geq \eta_{hard,m}$ is rejected. When $\eta > \eta_{soft,m}$ but below hard, it replaces R by

$$R_{eff} = \min(c_{max}, \eta/\eta_{soft,m})R \tag{32}$$

and recomputes the innovation covariance and gain. Joseph form is used:

$$P^+ = (I - KH)P^-(I - KH)^T + KR_{eff}K^T. \tag{33}$$

NIS is dimensionless. Default soft/hard thresholds for 1D, 2D and 3D are respectively $(6.635, 10.828)$, $(9.21, 13.816)$ and $(11.345, 16.266)$, with $c_{max} = 10$. The actual recorded profile is authoritative. Equality at the KF6 hard threshold rejects. A rejected or numerically failed update does not establish successful fusion.

5.4 Delayed observations and bounded replay

Trusted MCU sample time is retained. Otherwise the backend resolves measurement time as receive time minus the configured delay, rejecting underflow and future timestamps. The fixed-lag engine uses a 600 ms admission window, 144 inertial increments, 48 GNSS events, 160 barometer events and 8 checkpoints at a stride of 18 predictions. Its work bound is $144 + 2(48 + 160) = 560$ steps.

An observation inside an interval splits the navigation increment in proportion to time while retaining the original T for Equation (20). Replay restores filter state, covariance and relevant diagnostic/reacquisition state, sorts admitted events and repropagates to the present. History loss, overflow or replay failure has an explicit result; it is not an instruction to fall back to a current-state update.

Outage/reacquisition, covariance inflation and conditional reanchoring are project state machines beyond the basic Kalman equations. Receive-side loss/reset evidence is collected once; replay cannot manufacture a fresh physical reception or repeat task side effects. Their exact conditions remain in the owning outage/recovery contract and implementation, not in a universal Kalman-filter claim.

Implementation: S5, state order, original-interval noise, group observations, Cholesky/NIS and Joseph update. Implementation: S7, bounded fixed-lag replay and interval splitting.

6 ESKF15: right local error and residual biases

6.1 Nominal and error states

The nominal storage has 16 scalars,

$$(\boldsymbol{p}, \boldsymbol{v}, q_{nb}, \boldsymbol{b}_g, \boldsymbol{b}_a), \tag{34}$$

while the error state has 15 degrees of freedom:

$$\delta\boldsymbol{x} = [\delta\boldsymbol{p}^T, \delta\boldsymbol{v}^T, \delta\boldsymbol{\theta}^T, \delta\boldsymbol{b}_g^T, \delta\boldsymbol{b}_a^T]^T, \qquad q_{true} = q_{nb} \otimes \mathrm{Exp}(\delta\boldsymbol{\theta}). \tag{35}$$

The attitude error is a right-multiplicative Body rotation. Additive errors are true minus nominal. The biases are residuals after external calibration, not substitutes for the calibration result.

6.2 Body-input propagation

Firmware stores the actual calibrated means of two half intervals. With h their total duration, the implemented input increments are

$$\boldsymbol{a}_i = (\boldsymbol{\omega}_{m,i} - \boldsymbol{b}_g)h/2, \qquad \boldsymbol{u}_i = (\boldsymbol{f}_{m,i} - \boldsymbol{b}_a)h/2, \quad i = 1, 2. \tag{36}$$

Equations (6)–(11) then apply to this bias-corrected pair, using the ESKF nominal starting attitude and ESKF gravity. Nominal biases remain unchanged during deterministic prediction; their uncertainty grows through random-walk noise.

ESKF must consume Body inputs. Reusing already-rotated Pure INS ENU increments would lose the effect of attitude/bias corrections and risk repeated rotation or compensation. Its prediction rejects source/generation mismatch, noncontinuous time, excluded quality bits and $h \notin (0, 0.02]$ s. The stored endpoint duration and float duration must agree within 2×10^{-6} s.

6.3 Linearized dynamics and discretization

Let $\bar{\boldsymbol{\omega}} = (\boldsymbol{\omega}_{m,1} + \boldsymbol{\omega}_{m,2})/2 - \boldsymbol{b}_g$ and $\bar{\boldsymbol{f}} = (\boldsymbol{f}_{m,1} + \boldsymbol{f}_{m,2})/2 - \boldsymbol{b}_a$. In the error order of Equation (35),

$$F = \begin{bmatrix} 0 & I & 0 & 0 & 0 \\ 0 & 0 & -R[\bar{\boldsymbol{f}}]_\times & 0 & -R \\ 0 & 0 & -[\bar{\boldsymbol{\omega}}]_\times & -I & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{37}$$

For example, $R_{true} \simeq R(I + [\delta\boldsymbol{\theta}]_\times)$ gives $\delta(R\boldsymbol{f}) = -R[\boldsymbol{f}]_\times \delta\boldsymbol{\theta} - R\delta\boldsymbol{b}_a$. The right Body attitude error evolves as $\delta\dot{\boldsymbol{\theta}} = -[\bar{\boldsymbol{\omega}}]_\times \delta\boldsymbol{\theta} - \delta\boldsymbol{b}_g$. These signs must not be mixed with a left navigation-frame error model.

The implemented deterministic transition retains second order:

$$\Phi = I + Fh + \tfrac{1}{2}F^2h^2, \qquad P^- = \Phi P^+ \Phi^T + Q_d. \tag{38}$$

With the declared continuous amplitude densities, the isotropic firmware spectral matrix is

$$W = c_q \,\mathrm{diag}(0_3, n_a^2 I_3, n_g^2 I_3, n_{bg}^2 I_3, n_{ba}^2 I_3). \tag{39}$$

The process quality multiplier $c_q = 4$ when the input flags contain any `0x05` bit, otherwise $c_q = 1$; `0x72` excluded bits reject prediction. Densities are squared once. Units are given in the symbol table; sample standard deviation cannot be substituted without a sampling/noise model.

Noise propagation uses a *first-order* map even though deterministic Φ retains F^2:

$$\begin{aligned} Q_d &= \int_0^h (I + F\tau) W (I + F\tau)^T \, d\tau \\ &= hW + \tfrac{1}{2}h^2(FW + WF^T) + \tfrac{1}{3}h^3 FWF^T. \end{aligned} \tag{40}$$

Current declared defaults are $n_g = 0.003$, $n_a = 0.08$, $n_{bg} = 10^{-4}$ and $n_{ba} = 10^{-3}$ in their respective density units. FLP maps isotropic accelerometer noise through R and uses positive Simpson weights at $0, h/2, h$ for the same quadratic integrand. The mathematical covariance model agrees; float precision and actual half-duration rounding still matter. Firmware averages the two half means equally; FLP uses actual duration-weighted means for its reconstructed halves.

6.4 GNSS lever arm and actual observation matrices

Define $\boldsymbol{\omega} = \boldsymbol{\omega}_m - \boldsymbol{b}_g$ at measurement evaluation. The antenna predictions are

$$\hat{\boldsymbol{z}}_p = \boldsymbol{p} + R\boldsymbol{\ell}, \qquad \hat{\boldsymbol{z}}_v = \boldsymbol{v} + R(\boldsymbol{\omega} \times \boldsymbol{\ell}). \tag{41}$$

Their full three-axis Jacobians in error-state order are

$$H_p = [I \;\; 0 \;\; -R[\boldsymbol{\ell}]_\times \;\; 0 \;\; 0], \tag{42}$$

$$H_v = [0 \;\; I \;\; -R[\boldsymbol{\omega} \times \boldsymbol{\ell}]_\times \;\; R[\boldsymbol{\ell}]_\times \;\; 0]. \tag{43}$$

The five production groups select EN or U rows, as in SF6/KF6. Barometer prediction is relative up position with zero lever offset: $H_{baro} = [\boldsymbol{e}_U^T \;\; 0 \;\; 0 \;\; 0 \;\; 0]$. The positive gyro-bias block in H_v follows $(-\delta\boldsymbol{b}_g) \times \boldsymbol{\ell} = [\boldsymbol{\ell}]_\times \delta\boldsymbol{b}_g$.

Each supplied variance already includes backend quality/consistency adjustments; it is not sigma. The innovation, NIS and gain formulas have the same matrix meanings as KF6, but the firmware

innovation solve explicitly inverts a scalar or 2×2 system. Its separate covariance validator uses Cholesky. These are different operations.

ESKF rejects when $\eta > \eta_{hard,m}$ and soft-weights when $\eta > \eta_{soft,m}$ by $\min(c_{max}, \eta/\eta_{soft,m})$, then recomputes with inflated R. At exact hard equality, ESKF's comparison differs from KF6's $\geq$ comparison. Thresholds/noise/P0 follow the actual ESKF profile, not the KF6 settings. The declared 1D soft/hard pair is $(6.635, 10.828)$ and the 2D pair is $(9.21, 13.816)$; ESKF's default $c_{max} = 4$, whereas KF6's is 10.

The declared initial covariance diagonal follows $[\delta\boldsymbol{p}, \delta\boldsymbol{v}, \delta\boldsymbol{\theta}, \delta\boldsymbol{b}_g, \delta\boldsymbol{b}_a]$. Defaults are $(4, 4, 9)$ m^2, $(0.25, 0.25, 0.25)$ m^2/s^2, $0.030461742 I_3$ rad^2, $10^{-4} I_3$ rad^2/s^2 and $0.01 I_3$ m^2/s^4. These are variance entries already. Actual admitted initialization and its complete recorded covariance remain authoritative.

6.5 Injection, Joseph update and reset

After solving $\widehat{\delta\boldsymbol{x}} = K\boldsymbol{r}$, the candidate nominal correction is

$$\boldsymbol{p}^+ = \boldsymbol{p}^- + \widehat{\delta\boldsymbol{p}}, \qquad \boldsymbol{v}^+ = \boldsymbol{v}^- + \widehat{\delta\boldsymbol{v}}, \tag{44}$$

$$q^+ = \text{normalize}(q^- \otimes \text{Exp}(\widehat{\delta\boldsymbol{\theta}})), \tag{45}$$

$$\boldsymbol{b}_g^+ = \boldsymbol{b}_g^- + \widehat{\delta\boldsymbol{b}_g}, \qquad \boldsymbol{b}_a^+ = \boldsymbol{b}_a^- + \widehat{\delta\boldsymbol{b}_a}. \tag{46}$$

The error mean is recentered at zero. After the Joseph covariance update of Equation (33), the *full* covariance is transformed:

$$P^+ = JP_{Joseph}J^T, \qquad J = \text{diag}(I_3, I_3, J_r(\widehat{\delta\boldsymbol{\theta}}), I_3, I_3). \tag{47}$$

For $\alpha = \|\boldsymbol{a}\|$,

$$J_r(\boldsymbol{a}) = I - \frac{1 - \cos\alpha}{\alpha^2}[\boldsymbol{a}]_\times + \frac{\alpha - \sin\alpha}{\alpha^3}[\boldsymbol{a}]_\times^2. \tag{48}$$

The implementation uses coefficient series $1/2 - \alpha^2/24$ and $1/6 - \alpha^2/120$ for $\alpha < 10^{-3}$; the leading reset is $I - \frac{1}{2}[\boldsymbol{a}]_\times + \frac{1}{6}[\boldsymbol{a}]_\times^2$. This follows $\text{Exp}(\delta\boldsymbol{\theta}_{new}) = \text{Exp}(-\widehat{\delta\boldsymbol{\theta}})\,\text{Exp}(\delta\boldsymbol{\theta}_{old})$ linearized around the injected estimate. All attitude cross-covariances change under Equation (47), not only its attitude diagonal block.

The implementation limits attitude-correction norm to 0.35 rad, residual gyro-bias norm to 1 rad/s and residual acceleration-bias norm to 5 m/s^2. These are model-domain checks, not sensor accuracy specifications. Finite candidate state, nonzero finite gain norm and validated covariance are required before commit. Numerical/model errors must remain distinct from accepted fusion.

6.6 Bounded Body replay

ESKF retains 192 Body intervals, 208 events and a 600 ms window, with maximum $192+2(208) = 608$ replay steps. It must restore nominal state and covariance and reintegrate Body inputs after a correction, because corrected attitude and biases change subsequent ENU increments. Source, calibration generation, continuous epoch and capacity are part of admission. A reset is not crossed by replay.

Recorded initialization requires the matched ESKF initial state, complete initial covariance snapshot and Body-input identity. Offline initialization/default covariance is a different mode. Missing history or recorded evidence is an explicit failure/unavailable state, not an invitation to synthesize inputs.

Implementation: S6, nominal storage, right error, dynamics, noise, observation matrices and reset. Implementation: S8, ESKF backend parameters, timestamp/quality handling and bounded Body replay.

7 Preparation: calibration, alignment and initial state

Preparation is distinct from estimator feedback. Calibration establishes external correction and its provenance; alignment establishes the initial orientation; the admitted initial-state snapshot supplies the start boundary. An acknowledgement is not READY. Runtime command/lifecycle ownership remains authoritative.

7.1 External IMU correction and six-face calibration

For each selected Body axis, the applied correction is

$$f_{c,j} = (f_{raw,j} - b_{a,cal,j})s_{a,j}, \qquad \omega_{c,j} = (\omega_{raw,j} - b_{g,cal,j})s_{g,j}. \tag{49}$$

This is a per-axis bias/scale model, not a full misalignment matrix or temperature model. In six-face calibration, a_j^+ and a_j^- are the measured means on that axis's positive and negative faces:

$$b_{a,cal,j} = \tfrac{1}{2}(a_j^+ + a_j^-), \qquad s_{a,j} = \frac{2g}{a_j^+ - a_j^-}, \tag{50}$$

$$b_{g,cal,j} = \frac{1}{6}\sum_{face=1}^{6} \bar{\omega}_{face,j}, \qquad s_{g,j} = 1. \tag{51}$$

The denominator must be finite and greater than 10^{-6} m/s^2, and the scale must lie in the configured range. One-face and NONE procedures have their own implementation/lifecycle conditions; the six-face equation must not be described as the estimator's bias filter. ESKF subsequently subtracts its estimated *residual* biases from the corrected inputs of Equation (49).

Implementation: S9, six-face correction calculation and system correction application.

7.2 Alignment windows and yaw

Alignment accumulates a bounded admitted stationary window. Its inertial mean is $\bar{\boldsymbol{f}} = N^{-1}\sum \boldsymbol{f}_i$ and $\bar{\boldsymbol{\omega}} = N^{-1}\sum \boldsymbol{\omega}_i$ after the implemented validity, stationarity and timing checks. For hardware-quaternion averaging it forms a dyadic sum

$$M = \sum_i q_i q_i^T, \qquad q^{(r+1)} = \text{normalize}(Mq^{(r)}), \tag{52}$$

for the fixed iteration count in the owner. It seeds from the stored reference quaternion and resolves sign relative to that reference. This bounded numerical procedure must not be claimed to be an exact eigen-solution.

Gravity constrains tilt, but a single gravity vector does not observe yaw. Selected known-yaw or vector/external-attitude strategies supply the additional orientation constraint; unavailable/misaligned/collinear inputs must retain the selected strategy's rejection behavior. For known yaw, the code first rotates the selected signed Body axis into ENU, then computes

$$\psi_{current} = \text{atan2}(u_N, u_E), \qquad \Delta\psi = \text{wrap}_{[-\pi,\pi]}(\psi_{target} - \psi_{current}), \tag{53}$$

$$q_{aligned} = \text{normalize}(q_U(\Delta\psi) \otimes q_{source}). \tag{54}$$

This navigation-Up yaw adjustment is *left* multiplication, whereas Body inertial propagation and ESKF local injection multiply on the right. The projected axis must have usable horizontal norm. Replacing this with right multiplication generally changes the tilt-dependent result.

Calibration READY, alignment READY and successful START are separate boundaries. Hardware attitude used to establish/diagnose an initial state is not a continuing observation update in SF6/KF6. The source identity and calibration generation must remain coherent through initialization and prediction.

Implementation: S10, bounded inertial/quaternion means and known-yaw application.

7.3 Origin and initialization are not running corrections

GNSS origin and barometer reference are frozen from their admitted preparation windows. $\boldsymbol{p}_0 = 0$ refers to that local origin; it does not assert a perfect absolute position. Initial velocity uses only admitted components. The state quaternion is the accepted initial attitude. ESKF's complete initial P_0 belongs to its recorded initialization; generic KF6 diagonal floors cannot substitute for it. Missing origin/initialization evidence remains visible rather than silently rebasing a moving mission.

8 Quality windows, physical admission and health

These diagnostics are separate from coning/sculling and from the basic measurement update equations. Revision 3's covariance-based policies apply to the KF6/ESKF15 backend paths. SF6 uses its own physical/admission health contract in Section 4.

8.1 Receiver validity and satellite variance scale

Physical GNSS admission checks online/fix state, supported/valid fields, finite values, age and per-group capability. No fix is an ordinary rejection, not a fabricated valid zero. Low satellite count is quality degradation, not by itself proof of a bad position. When the count is valid,

$$c_{sv} = \begin{cases} 1, & N_{sv} \geq 6, \\ \min(4, 1 + \frac{1}{2}(6 - N_{sv})^2), & N_{sv} < 6. \end{cases} \tag{55}$$

An unavailable count gives this multiplier's neutral value 1; it does not certify healthy receiver quality. This multiplier acts on variance, not on sigma or on the SF6 gain. Other supported-field checks retain their own validity policy.

8.2 Two interleaved consistency windows

Current firmware configuration uses two 10 s windows offset by 5 s, with completed-evidence TTL 6 s. Let $\boldsymbol{p}_{EN}(t)$ and $\boldsymbol{v}_{EN}(t)$ be receiver horizontal position and velocity. For a covered window $[t_0, t_1]$, the diagnostic closure is

$$\boldsymbol{e} = \boldsymbol{p}_{EN}(t_1) - \boldsymbol{p}_{EN}(t_0) - \int_{t_0}^{t_1} \boldsymbol{v}_{EN}(t)\, dt, \qquad E = \|\boldsymbol{e}\|. \tag{56}$$

The code linearly interpolates endpoint positions and integrates linearly interpolated velocity exactly over each clipped sample segment. For fractions a, b of an interval with velocities $\boldsymbol{v}_i, \boldsymbol{v}_{i+1}$, the contribution is

$$\Delta\boldsymbol{d} = \left[\boldsymbol{v}_i + \tfrac{1}{2}(a+b)(\boldsymbol{v}_{i+1} - \boldsymbol{v}_i)\right] \Delta t_{clip}. \tag{57}$$

Finite valid position endpoints and continuous velocity evidence are needed. Source/generation change, a sequence gap, unusable velocity or excess timestamp gap anchors new windows and

invalidates the previous scale evidence. Duplicate/old evidence is not a new sample. There is no fabricated high-rate observation from a repeated low-rate receiver sample.

For configured error threshold $E_0 > 0$ and maximum $c_w^{max} \in [1, 4]$, a completed window yields

$$c_w = \min(c_w^{max}, \max(1, E^2/E_0^2)). \tag{58}$$

The current KF6 backend applies the completed-window scale to horizontal position variance. A stale/unavailable window returns neutral scale 1 and carries the unavailable evidence state; it does not prove closure consistency. This diagnostic is not position error against truth, RMSE against ground truth, a covariance proof or a navigation-accuracy certification.

The historical standalone `NavigationIntegrity` suspect/reject state machine remains a different implementation/history contract. It must not be presented as revision 3's active window algorithm merely because both use a position–velocity closure.

Implementation: S11, satellite variance scale, clipped integration, completed window and TTL. Implementation: S12, actual 10 s/5 s/6 s backend configuration and group scaling.

8.3 Fusion health is more than receipt

The health owner distinguishes receipt, physical validity, attempted update and *effective* successful fusion. Each of five groups retains the actual receive/sample/evaluation identity and last successful fusion time. Rejects or failed numerical updates must not refresh success. SF6 zero-gain observations do not provide an effective update even if admitted.

At the recorded source baseline, group supervision marks dead reckoning after 2 s without effective fusion and invalid after 10 s, measured from the prior success or initialization epoch. Required-group selection and IMU/model failures also participate. A fresh packet, ACK or successful file read is not equivalent to READY or a successful filter correction. Replay restores algorithm history while receive-side health evidence remains a single physical reception.

These timeouts are implemented health policy, not drift bounds. Continued inertial propagation, a finite covariance or a successful NIS test does not establish safe flight. Diagnostics need their own recorded timestamps, reason/result codes and unavailable status.

Implementation: S13, group supervision, successful-fusion ownership and model/IMU faults.

9 Recorded replay and evidence boundaries

Recorded analysis opens through the exact `.ssdecoder` and Descriptor. Source logs and decoder packages remain immutable. Explicit Offline What-if parameters must be labeled separately from recorded configuration; absent noise, initial state, validity or timing fields cannot be silently filled and called recorded.

Mathematical equivalence is distinct from numerical replay fidelity. Pure INS/KF6 currently lack an exact-validation reference on the ordinary reconstruction path; SF6 revision 2 is approximate when reconstructed, with unavailable status for missing required input. ESKF's recorded-parity path additionally checks actual Body inputs, complete initial state/covariance, timing, identities, measurement outcomes and state/covariance comparisons. A complete declared comparison can establish its recorded-data fidelity scope; it cannot establish real sensor truth or target hardware correctness. Missing covariance parts cannot be reported as compared successfully.

Host tests, source-graph checks, Power-of-Ten project gates, analyzer compilation and ARM linking establish only their executed revisions/configurations/scopes. Historical reports retain their original source identity, failed/skipped details and hashes. An unavailable hardware test

is not a pass; unresolved critical contracts remain NOT PROVEN. This mathematical reference adds no new MCU WCET, stack, electrical, hardware/flight or USB acceptance evidence.

10 Source index and maintenance map

Source keys refer to repository-relative owning files at the title-page commit. The root prefix for S1–S13 is `apps/SCG/plugins/builtin/`; paths below are relative to that prefix unless stated otherwise. Maintain equations when the owner/revision changes, and preserve the prior reference as a versioned historical document.

Key	Owning file(s) and evidence anchor
S1	`silverstar_algorithm_ins_coning2_sculling2/payload/Algorithm/INS/Coning2Sculling2/Src/ins_mechanization.c`: `Ins_ComputeSubIntervalIncrement`, coning/sculling, `InsMechanization_NavigationCompute`.
S2	`silverstar_algorithm_common/payload/Algorithm/Common/Src/attitude_frame.c`: Hamilton multiply, rotation-vector exponential and right Body propagation.
S3	`silverstar_algorithm_estimator_sf6/payload/Algorithm/Estimator/SF6/Inc/navigation_sf6.h` and `silverstar_algorithm_estimator_sf6/payload/Algorithm/Estimator/SF6/Src/navigation_sf6.c`: revision, state order, 500 ms admission, fixed-gain update.
S4	`silverstar_algorithm_estimator_sf6/payload/Algorithm/Estimator/SF6/Src/navigation_sf6_backend.c`: actual source/time handling, effective gain, unavailable covariance and health.
S5	`silverstar_algorithm_estimator_kf6/payload/Algorithm/Estimator/KF6/Src/navigation_kf.c`: `NavigationKf_PredictNoise`, `NavigationKf_PredictInterval`, vector updates, Joseph form and reacquisition. Parameter values: owning `silverstar_algorithm_estimator_kf6/plugin.json`.
S6	`silverstar_algorithm_estimator_eskf15/payload/Algorithm/Estimator/ESKF15/Src/navigation_eskf.c` and corresponding `Inc/navigation_eskf.h`: right error, nominal integration, linearization, noise, lever-arm Jacobians, reset and numerical limits. Parameter values: owning `silverstar_algorithm_estimator_eskf15/plugin.json`.
S7	`silverstar_algorithm_estimator_kf6/payload/Algorithm/Estimator/KF6/Inc/navigation_kf_replay.h` and corresponding `Src/navigation_kf_replay.c`: 600 ms window, capacities, checkpoint stride and original-interval splitting.
S8	`silverstar_algorithm_estimator_eskf15/payload/Algorithm/Estimator/ESKF15/Src/navigation_eskf_backend.c`; `Inc/navigation_eskf_replay.h` and `Src/navigation_eskf_replay.c`: profile initialization, source/generation/time admission and Body replay.
S9	`silverstar_algorithm_calibration/payload/Algorithm/Calibration/Src/imu_six_face_calibration.c`; `silverstar_core_0_1_0/payload/System/Calibration/Src/system_calibration_correction.c`: external bias/scale application and six-face model.
S10	`silverstar_algorithm_alignment_common/payload/Algorithm/Alignment/Common/Src/attitude_alignment.c`: bounded dyadic quaternion mean and known-yaw left adjustment. Selected vector/external-attitude strategies retain their own contracts.
S11	`silverstar_algorithm_common/payload/Algorithm/Common/Inc/navigation_quality.h` and corresponding `Src/navigation_quality.c`: revision 3, satellite scale and two-window closure.
S12	`silverstar_core_0_1_0/payload/APP/Src/estimator_task.c`: GNSS quality application, profile/initialization, actual window configuration and replay integration. ESKF's corresponding path is in S8.
S13	`silverstar_core_0_1_0/payload/System/Inc/system_navigation_health.h` and corresponding `Src/system_navigation_health.c`; `Src/system_gnss_quality.c`: timeouts, group evidence and physical GNSS admission.

Key	Owning file(s) and evidence anchor
S14	Root-relative FLP: `apps/FLP/src/silverstar_flp/plugins/algorithms/pure_ins/mechanization.py`; `sf6/current_state.py`, `kf6/filter.py`, `eskf15/filter.py` and `eskf15/verification.py` under the same algorithm directory. These own offline arithmetic, version routing and recorded-parity comparison boundaries.

Related current contracts: `docs/public/algorithms/NAVIGATION_IMPLEMENTATION.md`, `docs/FCCG/SF6_CURRENT_STATE.md`, `docs/public/SCG/KF6_FIXED_LAG_REPLAY.md`, `docs/public/SCG/KF6_OUTAGE_RECOVERY.md`, `docs/public/SCG/ALGORITHM_PARAMETERS.md`, `docs/public/SCG/NAVIGATION_CONTRACT.md` and `docs/public/FLP/Field_Log_Replay.md`. Older pages bearing a historical date/version remain historical evidence rather than a current product-version authority.